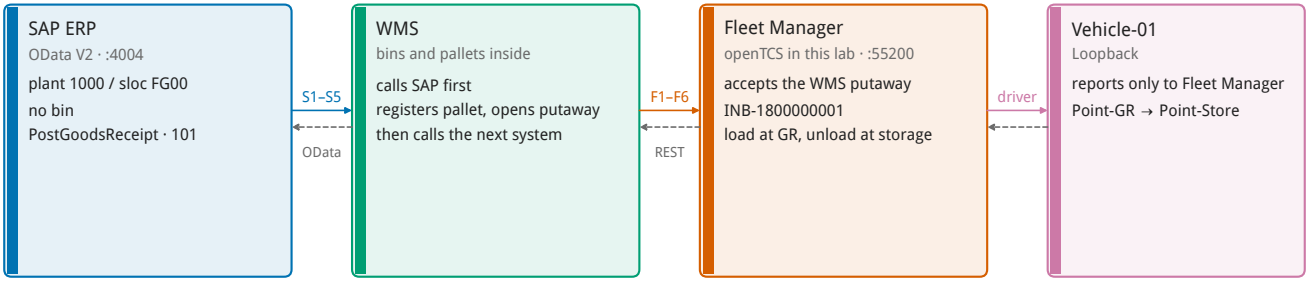
a One serial chain: SAP → WMS → Fleet Manager → vehicle. Bins stay inside the WMS.



Order: ① WMS reads SAP ② WMS putaway ③ WMS calls Fleet Manager ④ vehicle ⑤ WMS posts goods receipt

b

ID	Method	Path	Purpose	Phase
SAP ERP · {INB} = /sap/opu/odata/sap/API_INBOUND_DELIVERY_SRV;v=0002				
{PROD} = API_PRODUCT_SRV, {OUT} = API_OUTBOUND_DELIVERY_SRV;v=0002; posting needs X-CSRF-Token and If-Match				
S1	GET	{INB}/$metadata	readiness, metadata	①
S2	GET	{INB}/A_InbDeliveryHeader?$filter=…Status eq 'A'	poll open inbound deliveries	②
S3	GET	{INB}/A_InbDeliveryHeader('1800000001')	read status and ETag	⑥
S4	HEAD	{INB}/ X-CSRF-Token: Fetch	fetch CSRF token	⑥
S5	POST	{INB}/PostGoodsReceipt?DeliveryDocument='1800000001'	post goods receipt, 101	⑥
S6	GET	{PROD}/A_Product A_ProductStorageLocation	sync material and storage location	①
S7	POST	{OUT}/PostGoodsIssue?DeliveryDocument='8000000001'	goods issue, not automatic here	outbound
Inside the WMS · locations and pallets, not another system · OpenWMS Common :8120				
C1	GET	/actuator/info	readiness	①
C2	GET POST	/v1/locations?locationId=…	find / create location	①
C3	GET POST	/v1/transport-units?bk=…[&actualLocation=…&tut=PALLET]	find / create pallet	①
C4	PATCH	/v1/transport-units?bk=…&newLocation=…	move pallet	③ ⑥
Inside the WMS · putaway orders, not another system · OpenWMS TMS :8131				
T1	GET	/actuator/info	readiness	①
T2	POST	/v1/transport-orders	create order, 201 + Location	③
T3	GET	/v1/transport-orders/{pKey}	read order state	③ ⑥
T4	POST	/v1/transport-orders/{pKey}?state=STARTED|FINISHED|ONFAILURE|CANCELED	change order state	③ ⑥
Fleet Manager · openTCS in this lab · http://opentcs:55200				
F1	GET	/v1/kernel/version	readiness	①
F2	GET PUT	/v1/plantModel	read / upload plant model	①
F3	PUT	/v1/vehicles/Vehicle-01/commAdapter/…	attach Loopback, allow orders	①
F4	POST	/v1/transportOrders/INB-{DeliveryDocument}	putaway: load at GR, unload at storage	④
F5	GET	/v1/transportOrders/INB-{DeliveryDocument}	read order state	⑤
F6	POST	/v1/transportOrders/INB-{DeliveryDocument}/withdrawal?immediate=false	withdraw on failure	failure
WMS · http://wms:8090				
W1	GET	/status /health	delivery links, progress, SAP calls	observe

Phases follow the call order: ① init ② WMS queries SAP ③ putaway inside WMS ④ WMS calls Fleet Manager ⑤ vehicle ⑥ WMS posts to SAP. The vehicle talks only to Fleet Manager.